



The **AiEdge** ACADEMY

The Train Fine-Tune Deploy LLMs Bootcamp

Myself



Damien Benveniste

- Physics PhD
- 15 years of ML experience
- 10 years in the industry
- Applied ML to Ads Ranking, Market Research, Financial Advising, Healthcare, Credit worthiness, E-commerce, cloud computing
- Now entrepreneur – educator

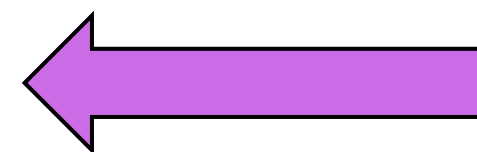
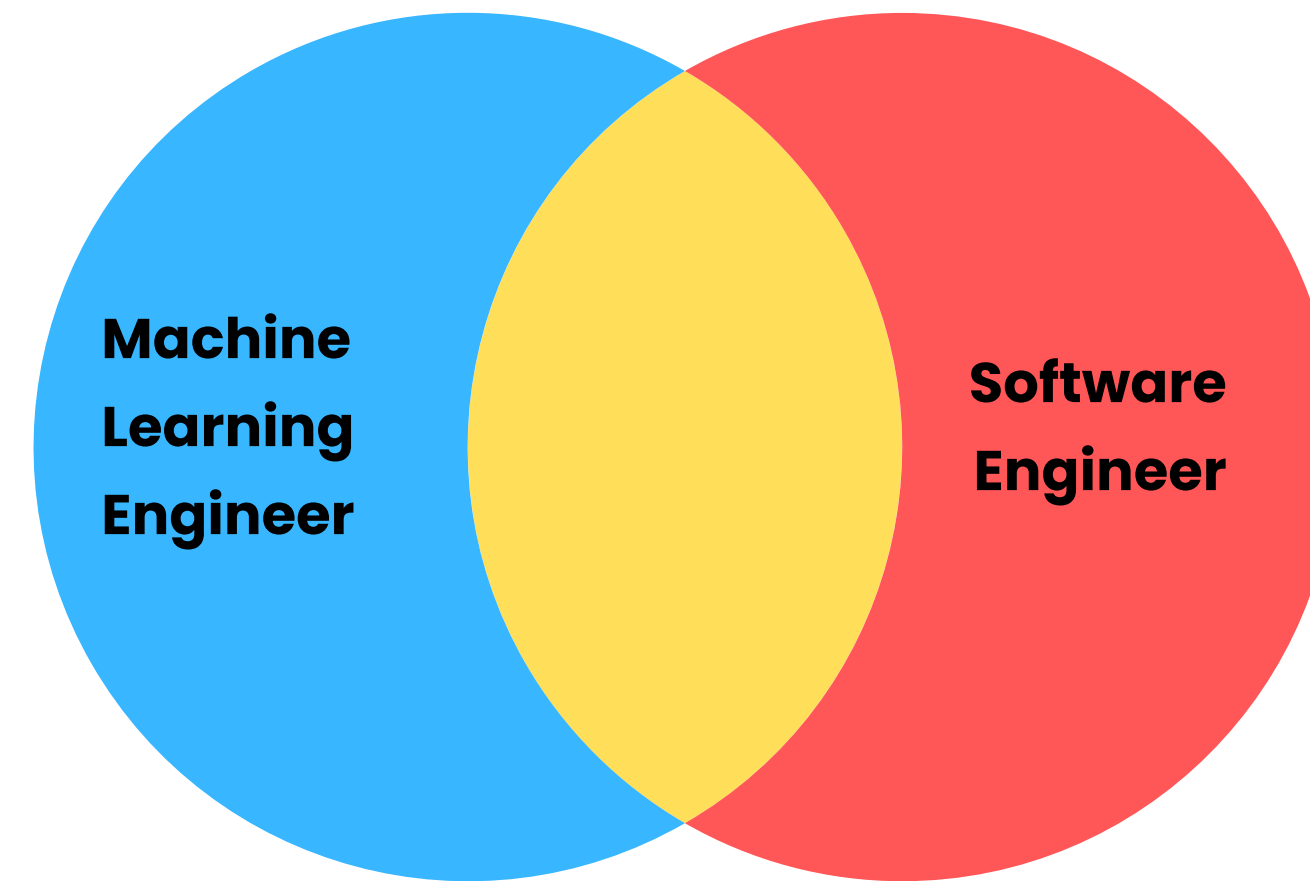
Please, fill the introduction form “Tell me more about you!”

NLP Machine Learning Engineer

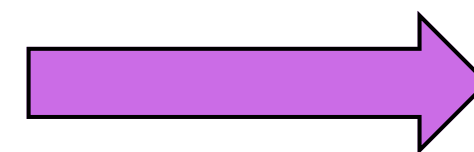
Train, Deploy, fine-tune models

Responsibilities: Specialize in training or customizing pre-trained NLP models. This involves data preprocessing, model fine-tuning, evaluation, and deployment of models to production environments.

Skills: PyTorch, HuggingFace Packages, ...



Deploy / fine-tune models



Use orchestrated frameworks and API

Use orchestrated frameworks and API

Responsibilities: Focus on developing NLP applications and services by leveraging high-level API frameworks like Langchain or LlamaIndex. This includes integrating NLP capabilities into applications.

Skills: Prompt Engineering, LangChain, LlamaIndex, RESTful APIs, Python, web development frameworks (FastAPI, Django), full-stack development, Basic knowledge of NLP.

The Curriculum

- The Transformer Architecture (1 week)
- Training LLMs to Follow Instruction (1 week)
- How to Scale Model Training (1 week)
- How to Fine-Tune LLMs (1 week)
- How to Deploy LLMs (1 week)
- Building the Application Layer (1 week)

Learning the Foundations for the Future



Future Courses to come:

- Orchestrated LLM Applications
- Introduction to Data Science
- The Mathematics of Machine Learning
- How to deploy ML models
- MLOps
- Recommender systems
- Software Engineering Best Practices for ML
- ...

The Homework



The most important piece of the bootcamp!

- Work together but submit your own work!
- They are meant to be long and hard! Do your best!
- Going beyond the lectures
- If you ask a question, make sure you investigate before
- You will be graded on the homework

The Homework

- **Project 1:** Implementing from scratch The sparse attention mechanisms, SliGLU, RMSNorm, MoE, and Rope embedding in PyTorch
- **Project 2:** Fine-tuning an LLM with PPO vs DPO vs ORPO using the PEFT package.
- **Project 3:** Train an LLM in a distributed manner with the Accelerate package in AWS SageMaker with the Zero Redundancy Optimizer Strategy.
- **Project 4:** Fine-tuning a model with QLoRA to increase the context size.
- **Project 5:** Deploying a scalable LLM application API with streaming, KV-caching, Continuous batching, and text generation layer capabilities.
- **Project 6:** Deploying an RAG application using LangChain, FastAPI, and LangServe.

The Tools

- **Discord server:** I created a Discord server for our Bootcamp. Follow this link to sign into it: <https://discord.gg/fMT8pBVKpG>
- **Comments:** Make sure to leave comments at the bottom of each lecture if you have any questions.
- **Jupyter Notebook:** We are going to use Jupyter notebooks for quick prototyping. Make sure to install it: <https://www.anaconda.com/download/>
- **VSCode:** We are going to use VSCode for software development. Feel free to use your preferred IDE, though. Make sure to install it: <https://code.visualstudio.com/>